T3 group factors: EFT_2_after_S1_S2

Active theory: SMEFT with the Weinberg operator C_5. No T3 heavy field remains dynamical.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q}\, Y_u\, \widetilde{H}\, u + \text{h.c.}$
Y_d	$-\bar{Q}\, Y_d\, H\, d + \text{h.c.}$
Y_e	$-\bar{L}\, Y_e\, H\, e + \text{h.c.}$
T	$\mathrm{Tr}\left(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d\right)$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
C_5^{ij}	$(L_i^T C L_j)\, H H$

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-A	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	S1, S2	none
T3-A	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	S1, S2	none
T3-A	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	S1, S2	none
T3-B	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	S1, S2	none
T3-B	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	S1, S2	none
T3-B	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	S1, S2	none
T3-C	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	S1, S2	none
T3-C	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	S1, S2	none
T3-C	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	S1, S2	none
T3-D	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	S1, S2	none
T3-D	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	S1, S2	none
T3-D	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	S1, S2	none
T3-E	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	S1, S2	none
T3-E	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	S1, S2	none
T3-E	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	S1, S2	none
T3-E	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	S1, S2	none

Decoupling statement

All dependence on the original S_1, S_2 and F representations is contained in the matched boundary value $C_5(\mu_{\rm th})$. The one-loop running below the final threshold is common to all T3 classes.

β_{C_5}

Run block 1: T3-A–T3-A

$\mathcal{S}$	T3 − A ($\alpha=-4$)	T3 − A ($\alpha=-2$)	T3 − A ($\alpha=0$)

$\lambda_1\, C_5$	2	2	2
$g_2^2\, C_5$	-3	-3	-3
$T\, C_5$	2	2	2
$(Y_e^\dagger Y_e)^T C_5$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$C_5 (Y_e^\dagger Y_e)$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

Run block 2: T3-B–T3-B

$\mathcal{S}$	T3 − B ($\alpha = -3$)	T3 − B ($\alpha = -1$)	T3 − B ($\alpha = 1$)
$\lambda_1\, C_5$	2	2	2
$g_2^2\, C_5$	-3	-3	-3
$T\, C_5$	2	2	2
$(Y_e^\dagger Y_e)^T C_5$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$C_5 (Y_e^\dagger Y_e)$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

Run block 3: T3-C–T3-C

$\mathcal{S}$	T3 − C ($\alpha = -3$)	T3 − C ($\alpha = -1$)	T3 − C ($\alpha = 1$)
$\lambda_1\, C_5$	2	2	2
$g_2^2\, C_5$	-3	-3	-3
$T\, C_5$	2	2	2
$(Y_e^\dagger Y_e)^T C_5$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$C_5 (Y_e^\dagger Y_e)$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

Run block 4: T3-D–T3-D

$\mathcal{S}$	T3 − D ($\alpha = -2$)	T3 − D ($\alpha = 0$)	T3 − D ($\alpha = 2$)
$\lambda_1\, C_5$	2	2	2
$g_2^2\, C_5$	-3	-3	-3
$T\, C_5$	2	2	2
$(Y_e^\dagger Y_e)^T C_5$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$C_5 (Y_e^\dagger Y_e)$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

Run block 5: T3-E–T3-E

$\mathcal{S}$	T3 − E ($\alpha = -4$)	T3 − E ($\alpha = -2$)	T3 − E ($\alpha = 0$)
$\lambda_1\, C_5$	2	2	2
$g_2^2\, C_5$	-3	-3	-3
$T\, C_5$	2	2	2
$(Y_e^\dagger Y_e)^T C_5$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$C_5 (Y_e^\dagger Y_e)$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

Run block 6: T3-E–T3-E

$\mathcal{S}$	T3 − E ($\alpha = 2$)
$\lambda_1\, C_5$	2
$g_2^2\, C_5$	-3
$T\, C_5$	2
$(Y_e^\dagger Y_e)^T C_5$	$-\frac{3}{2}$
$C_5 (Y_e^\dagger Y_e)$	$-\frac{3}{2}$

Flavor form

$$16\pi^2 \frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \tag{1}$$

$$- \frac{3}{2}\left[(Y_e^\dagger Y_e)^T C_5 + C_5(Y_e^\dagger Y_e)\right]. \tag{2}$$